gpeyt1-16 · Cyclic stress ratio

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A bridge abutment site is underlain by loose to medium dense clean sand. Water table at 6 ft. γ = 112 pcf above and γsat = 125 pcf below. The design earthquake has Mw = 7.0 and a site-adjusted peak ground acceleration amax = 0.32g (PGA on rock was 0.24g). Evaluate the sand at 24 ft depth using the Seed–Idriss simplified procedure with the mean rd relation.

Most nearly, the CSR at 24 ft is:

A. 0.338
B. 0.196
C. 0.319
D. 0.239

Given equations / data

> CSR = 0.65 (amax/g)(σvo/σ′vo) rd; rd = 1.0 − 0.00233z for z < 30 ft (z in ft); γw = 62.4 pcf

Worked solution

1. σvo = 6(112) + 18(125) = 2,922 psf
2. u = 18(62.4) = 1,123 psf → σ′vo = 1,799 psf
3. rd = 1 − 0.00233(24) = 0.9441
4. CSR = 0.65 × 0.32 × (2,922/1,799) × 0.9441 = 0.319

Final answer

0.319

Why the other values are wrong

- **0.338** — omitted rd
- **0.196** — omitted σvo/σ′vo
- **0.239** — used rock PGA instead of site amax

References (Jayce's library)

- FHWA-NHI-11-032 (GEC 3) Eq. 6-16, §6.4 Simplified Procedure; rd relation (Liao & Whitman) in GEC 3

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=Hg3MTAvoJWw, https://www.youtube.com/watch?v=f9PIvbPStrU (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.